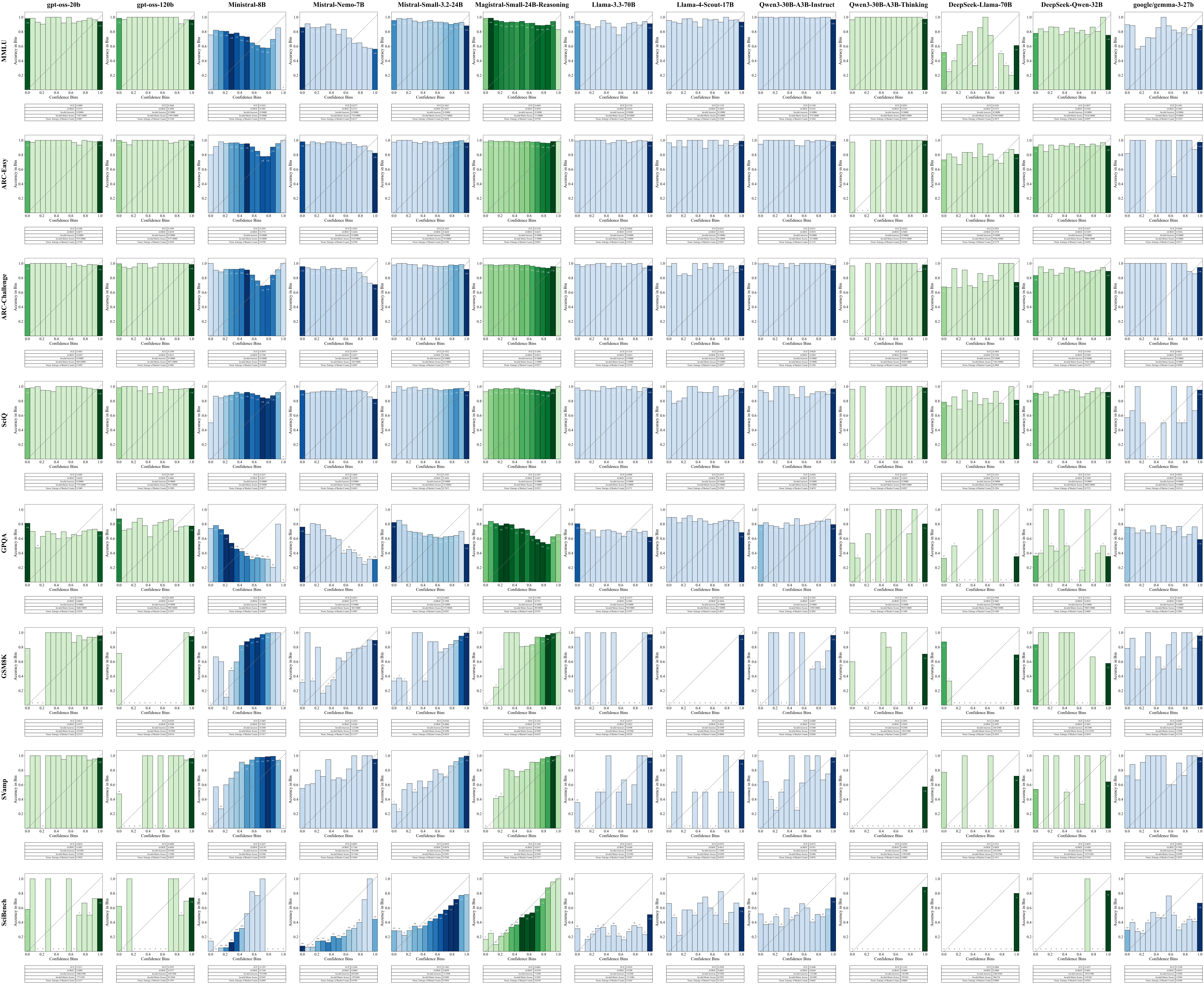
Calibration Plots for Uncertainty Metric P(True)



MMLU

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.4909	0.3668	0.3422	0.6277	0.3667	0.4469	0.3729	0.1139	0.1394	0.0591	0.4182	0.4057	0.1491
AUROC	0.3372	0.3898	0.3985	0.3151	0.3837	0.3858	0.4510	0.4459	0.4182	0.5107	0.5372	0.4769	0.5483
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	1785/10000	1594/10000	0/10000	374/10000	121/10000	11/10000	20/10000	0/10000	375/10000	6803/10000	8744/10000	7614/10000	0/10000
Norm. Entropy of Bucket Counts	0.4067	0.3160	0.9190	0.6117	0.8858	0.9748	0.4321	0.2180	0.3206	0.0925	0.3073	0.4997	0.1453

ARC-Easy

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.3288	0.1995	0.2924	0.5443	0.1435	0.3230	0.0926	0.0231	0.0331	0.0162	0.2923	0.3427	0.0280
AUROC	0.4875	0.4678	0.3774	0.2644	0.4628	0.4623	0.4742	0.5636	0.5074	0.5069	0.5370	0.5189	0.5364
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	879/10000	875/10000	0/10000	618/10000	34/10000	2/10000	3/10000	0/10000	22/10000	5474/10000	6566/10000	5008/10000	0/10000
Norm. Entropy of Bucket Counts	0.3769	0.2650	0.8780	0.6780	0.6788	0.8967	0.2014	0.0587	0.1119	0.0298	0.3179	0.6449	0.0117

ARC-Challenge

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.3463	0.2398	0.3059	0.5879	0.1922	0.3386	0.1310	0.0467	0.0620	0.0346	0.3405	0.3564	0.0621
AUROC	0.4295	0.4413	0.3700	0.2657	0.3686	0.4014	0.4633	0.5156	0.4494	0.5018	0.5196	0.5500	0.5037
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	969/10000	887/10000	0/10000	520/10000	42/10000	3/10000	2/10000	0/10000	8/10000	5850/10000	7183/10000	5762/10000	0/10000
Norm. Entropy of Bucket Counts	0.3699	0.2882	0.8940	0.6487	0.7373	0.9227	0.2524	0.0927	0.1436	0.0469	0.3004	0.6372	0.0289

SciQ

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.2903	0.1907	0.2637	0.5069	0.1922	0.3587	0.0988	0.0359	0.0428	0.0237	0.3333	0.3341	0.0501
AUROC	0.4426	0.4884	0.4617	0.3470	0.4426	0.4567	0.5181	0.5819	0.5319	0.5076	0.5178	0.5209	0.5364
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	779/10000	951/10000	0/10000	927/10000	32/10000	2/10000	4/10000	0/10000	0/10000	5099/10000	6038/10000	4806/10000	0/10000
Norm. Entropy of Bucket Counts	0.3480	0.2490	0.8677	0.6693	0.7813	0.9322	0.2171	0.0780	0.0670	0.0207	0.3204	0.5733	0.0114

GPQA

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.5394	0.4902	0.3176	0.6813	0.4469	0.3590	0.5273	0.3434	0.3503	0.2149	0.5994	0.5411	0.4628
AUROC	0.4050	0.4167	0.3205	0.2779	0.3598	0.3731	0.3885	0.3638	0.4937	0.5531	0.4962	0.4935	0.3869
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	2686/10000	990/10000	1/10000	382/10000	272/10000	80/10000	90/10000	3/10000	2250/10000	9653/10000	9508/10000	9485/10000	3/10000
Norm. Entropy of Bucket Counts	0.4230	0.3297	0.8697	0.5087	0.8391	0.9791	0.5597	0.4011	0.5805	0.1496	0.1840	0.4069	0.3801

GSM8K

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0616	0.0528	0.3085	0.1022	0.0226	0.1242	0.0333	0.0340	0.0400	0.2992	0.4988	0.5647	0.0899
AUROC	0.5472	0.5590	0.7622	0.6346	0.8006	0.7397	0.5022	0.4942	0.5363	0.5055	0.3992	0.3691	0.6587
Invalid Answers	15/2500	2/2500	8/2500	12/2500	0/2500	0/2500	18/2500	0/2500	0/2500	0/2500	244/2500	30/2500	0/2500
Invalid Metric Scores	54/2485	10/2498	1/2492	33/2488	0/2500	0/2500	0/2500	0/2500	6/2500	1203/2500	2072/2256	2131/2470	2/2500
Norm. Entropy of Bucket Counts	0.2315	0.0134	0.7417	0.3137	0.4618	0.6187	0.0229	0.0000	0.0268	0.0457	0.2655	0.3018	0.1374

SVamp

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0624	0.0408	0.2657	0.0967	0.0410	0.1448	0.0333	0.0570	0.0372	0.4285	0.3321	0.4029	0.0892
AUROC	0.5491	0.6050	0.8170	0.7146	0.8978	0.8517	0.6560	0.6012	0.6761	0.4993	0.4828	0.5480	0.7943
Invalid Answers	18/2500	0/2500	54/2500	39/2500	9/2500	6/2500	8/2500	1/2500	14/2500	1/2500	102/2500	19/2500	1/2500
Invalid Metric Scores	52/2482	1/2500	0/2446	81/2461	0/2491	0/2494	25/2492	0/2499	0/2486	1199/2499	2178/2398	2155/2481	0/2499
Norm. Entropy of Bucket Counts	0.2829	0.0475	0.8358	0.2644	0.5548	0.7377	0.0567	0.0254	0.0876	0.0000	0.1411	0.3105	0.2075

SciBench

	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Magistral-Small-24B-Reasoning	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.3211	0.2935	0.1297	0.2268	0.1302	0.0605	0.4524	0.3920	0.2600	0.1143	0.2000	0.1657	0.3190
AUROC	0.5809	0.5577	0.7348	0.6860	0.6859	0.6794	0.5700	0.6061	0.6644	0.5000	0.5000	0.4881	0.6933
Invalid Answers	1080/2500	456/2500	55/2500	60/2500	11/2500	7/2500	14/2500	9/2500	12/2500	136/500	2284/2500	1932/2500	6/2500
Invalid Metric Scores	177/1420	81/2044	0/2445	129/2440	0/2489	5/2493	111/2486	0/2491	10/2488	329/364	206/216	518/568	4/2494
Norm. Entropy of Bucket Counts	0.2317	0.1391	0.6094	0.8705	0.9364	0.8330	0.3443	0.1651	0.6042	0.0000	0.0000	0.0362	0.4586